



UNIVERSITY OF MORATUWA
Faculty of Information Technology

B.Sc. (Hons) in Information Technology
Level 1 – Semester 2 Examination

IN1500 Data Communication

Time Allowed: 3 hours

July 2022

ADDITIONAL MATERIAL

None

INSTRUCTIONS TO CANDIDATES

1. This paper contains six (6) pages (including this page).
2. This paper comprises two (2) sections, namely **Section A** and **Section B**.
3. Section A contains five (5) short answer questions and Section B contains three (3) essay-type questions.
4. The total marks obtainable for this examination is 100. The marks assigned for each Question & Section thereof are included in brackets.
5. This examination accounts for 60% of the module assessment.
6. This is a closed book examination.
7. All the workings need to be shown very clearly.
8. Answer **ALL Questions in both Sections A & B**.
9. Answer Section A and B in **separate** booklets and tie them together.
10. Non-programmable scientific calculators are allowed.

Continued...

Section A

Answer all questions, clearly showing the workings.

(All complete and correct answers will carry 5 Marks each)

- Q1:** In a certain data communication channel, data is transferred using 16-QAM modulation with eight different phase angles and with two different frequencies. If the baud rate of data transmission is 2000, what is the transmission speed in bps?
- Q2:** A frame of 150 bytes is sent over a communication channel. Compute the total number of bits transmitted for the following cases.
- Asynchronous transmission with one start bit and two stop bits at each byte, and one start of frame and one end of frame characters (1 byte).
 - Synchronous transmission with one start of the frame and one end of the frame characters (1 byte).
- Q3:** A 100km long fiber optical fiber system consists of four segments of lengths 10, 20, 30, and 40 km respectively, joined by two repeaters at points B and D and a splice at point C as shown in Figure A3.

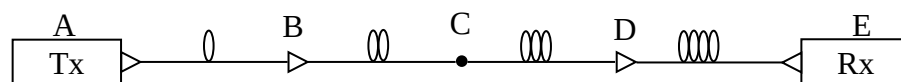


Figure A3

The following are the gain/loss values of different components in the system.

Repeater gains – 25 dB

Splice loss – 5 dB

Cable loss – 0.5 dB/km

Calculate the received signal strength in milliwatts at the receiver, if the transmit signal power is 1 W.

- Q4:** Compute the bandwidth of the channel in bps used to transmit a video stream with the following parameters.

No. of frames per second – 25

Size of a frame – 13”x8”

Resolution – 800x600 dpi

No. of colours per pixel supported – 65,536

Q5: Sketch (draw) the resulting signal patterns for the data stream 11000101011 using the following encoding schemes.

- a.) Alternate Mark Inversion (AMI) Encoding
- b.) Manchester Encoding
- c.) Differential Manchester Encoding
- d.) Non-Return to Zero – L (NRZ–L) Encoding
- e.) Non-Return to Zero – I (NRZ–I) Encoding

End of Section A

Section B

Answer all questions.

Question One

- (a) Name the five different types of topologies that are used to connect communication devices (or computers). [5 Marks]
- (b) Briefly describe the three different types of transmission modes used to communicate between two different devices. [6 Marks]
- (c) With the aid of suitable diagrams, explain the effect of noise on parallel transmission lines and describe how it can be minimized. [4 Marks]
- (d) A computer transmits 10,000 bytes of data over a full-duplex point-to-point link with a bandwidth of 100 Mbps with the following parameters.
- Size of the frame: 2048 bytes
- Overhead per frame: 40 bytes
- (i) What is the total time taken to transmit this data? [6 Marks]
- (ii) What is the effective throughput of the data transmission? [4 Marks]

Question Two

- (a) Giving at least three specific characteristics, compare and contrast digital signals against analog signals used in electronic communication. [6 Marks]
- (b) With the aid of suitable diagrams, describe the analog-to-digital conversion process and explain each stage sufficiently. [4 Marks]
- (c) An analog input signal has a maximum bandwidth of 8 kHz and varies between +5 and -5v. This signal is sampled, quantized into 256 levels, and then encoded using Pulse Code Modulation (PCM). Compute the following:
- (i) What is the minimum sampling rate required for converting this analog signal to a digital signal avoiding aliasing? [3 Marks]

- (ii) What are the corresponding quantized values and resulting bit patterns for the input signal values of -2.5V and $+1.30\text{V}$, assuming linear quantization?

[8 Marks]

- (iii) What is the resultant bit rate of the digital stream?

[4 Marks]

Question Three

- (a) With the aid of a suitable example, explain the term Equivalent Isotropic Radiated Power (EIRP).

[5 Marks]

- (b) Derive the formula for Free Space Loss (FSL) in wireless communication.

[5 Marks]

- (c) Figure Q3 shows the block diagram of a wireless communication system. The parameters of the communication system are as follows:

Transmitter power at the end of the amplifier: 2 W

Maximum distance between the transmitter and the receiver antennas: 20 km

Frequency of the signal used in the communication: 2.4 GHz.

Filter losses at Tx and Rx ends: 0.5 dB (each)

Feedline losses at Tx and Rx ends: 2 dB (each)

Tx antenna gain: 20 dBi

Rx antenna gain: 15 dBi

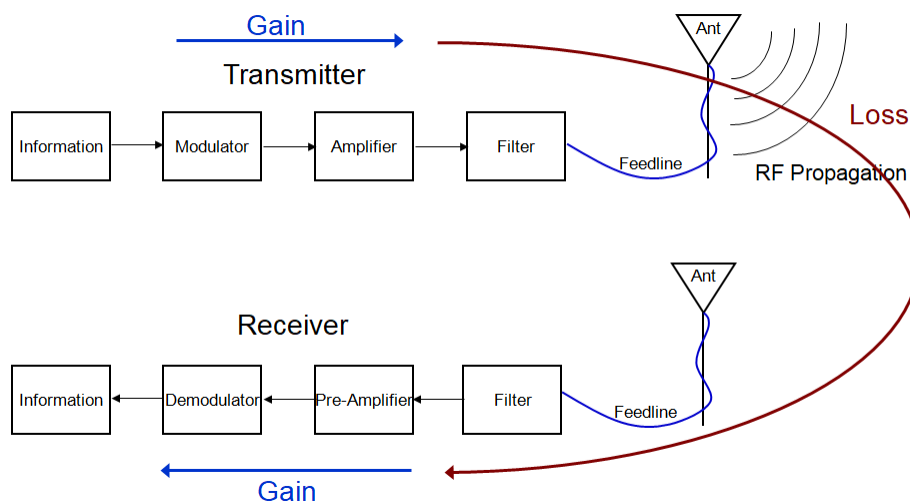


Figure Q3

Calculate the following:

(i) The EIRP at the transmitter in milliwatts (mW).

[5 Marks]

(ii) The total free space path loss in dB.

[5 Marks]

(iii) Received power at the receiver pre-amplifier in nanowatts (nW).

[5 Marks]

End of Section B

End of Paper